

MAS0257: Applied SDEs.

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## §1. Introduction

Example:  $X_t \in [0,1]$ , proportion  
of a population with a given  
genetic mutation at time  $t$ .  
 $t \in [0, \infty)$

Evolution depends on current  
state  $X_t$ , and random "noise".

loosely,  $\frac{dX_t}{dt} = \mu(X_t) + \sigma(X_t) \times \text{"noise"}$   
(1)

$\sigma \equiv 0 \Rightarrow$  "classical" ODE  
 $\rightarrow$  How to model "noise"?

## Typical Questions:

- How to model noise?  
What statistical properties?
  - What can we say about  
distribution of  $X_t$ ?
  - Path properties. E.g.  
probability population ever  
exceeds given level?
-

Properties of noise:

Discrete:  $\Delta t = N^{-1}$ ,  $t_k = k \Delta t$

$$X_1 = X_0 + \sum_{k=0}^{N-1} (\mu(X_k) \Delta t + \sigma(X_k) \tilde{z}_k)$$

Expect:  $\tilde{z}_k$ 's independent & identically distributed noise

Qn: What dist<sup>n</sup> for  $\tilde{z}$ ?  
how does it depend on  $\Delta t$ ?

$$\mu(x) = 0, \sigma(x) = 1, N=1$$

$$\text{Then } X_1 = X_0 + \sum_{k=0}^{N-1} \tilde{z}_k$$

CLT  $\Rightarrow$  This has a random limit

$$\text{if } \mathbb{E}[\tilde{z}_k] = \frac{\nu}{N}$$

$$\text{Var}(\tilde{z}_k) = \frac{\gamma^2}{N}$$

$$\text{Then } X_1 - X_0 \sim N(\nu, \gamma^2)$$

$$\text{More generally } X_t - X_s \sim N(\nu(t-s), \gamma^2(t-s))$$

and independence of  $\tilde{z}_i$ 's

$\Rightarrow$  given  $0 \leq t_0 \leq t_1 \leq \dots \leq t_k$

$$X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \dots$$

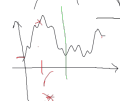
are independent.

Properties of  $X_t$ :

(P1) Normal increments

$$X_t - X_s \sim N(\nu(t-s), \gamma^2(t-s))$$

(P2) Independent increments.



(P3) Assume  $t \mapsto X_t$  is continuous.

Def<sup>n</sup> -  $X_t, X'_t$  are modifications if

$$\text{if } \forall t \quad \mathbb{P}(X_t = X'_t) = 1$$

-  $X_t, X'_t$  are indistinguishable

$$\text{if } \mathbb{P}(X_t(\omega) = X'_t(\omega), \forall t \geq 0) = 1$$

Thm (Kolmogorov's Continuity Criterion)

If  $X_t$  is a real-valued process

and  $\forall T > 0, \exists \alpha, \beta, \gamma > 0$  s.t.

$$\mathbb{E} |X_{t+h} - X_t|^\alpha \leq \gamma h^{1+\beta} \quad \forall t, \forall 0 \leq h \leq T$$

then  $X$  has a continuous modification.

NB (P1)(P2)  $\Rightarrow$  this holds

## §2. Brownian Motion

### §2.1 Definition & Basic Properties.

Def<sup>n</sup>:  $(W_t)_{t \geq 0}$  is a (standard) Brownian motion starting at  $x \in \mathbb{R}$  if,

- (P0)  $W_0 = x$
- (P1)  $W_t - W_s \sim N(0, t-s)$
- (P2) Independent increments
- (P3)  $t \mapsto W_t$  is continuous almost surely.

Properties i) B.M. is Strong Markov.

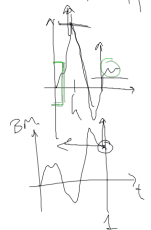
ii) BM is a martingale

iii) if  $\alpha \neq 0$   $(\alpha^{-1} W_{\alpha^2 t})_{t \geq 0}$  is B.M.

iv)  $(W_{t+h} - W_h)$  is B.M.

v)  $(W_1 - W_{1-t})_{t \in [0,1]}$  is B.M.

Thm BM exists & is unique



BM is scaling limit of random walks:

$$Z_n \text{ i.i.d } \mathbb{E} Z_k = 0, \text{Var}(Z_k) = 1$$

Set  $Y_n = \sum_{k=1}^n Z_k$

Random Walk

$$Y(t) = Y_{[t]} + (t - [t]) \times \left( \frac{Y_{[t]+1} - Y_{[t]}}{1 - [t]} \right)$$

Then  $Y(t)$  is continuous on  $[0, \infty)$

$$Y^N(t) := \frac{1}{\sqrt{N}} Y(Nt)$$

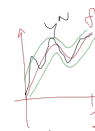
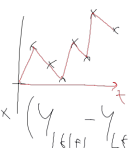
$\Rightarrow$  Thm (Donsker's Invariance Principle)

$Y^N$  converges to B.M.:

$\forall f \in C[0,1], \varepsilon > 0$

$$\mathbb{P} \left( \sup_{t \in [0,1]} |Y^N(t) - f(t)| < \varepsilon \right)$$

$$\xrightarrow{N \rightarrow \infty} \mathbb{P} \left( \sup_{t \in [0,1]} |W_t - f(t)| < \varepsilon \right)$$





"Formally"

have in (1)

$$\sum_{s \leq t} \text{noise}_s \rightarrow \int_0^t \text{noise}_s \approx W_t$$

so (1) says:

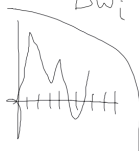
$$\frac{dX_t}{dt} = \mu(X_t) + \sigma(X_t) \cdot \dot{W}_t$$

But  $W_t$  is not differentiable.  
Need to interpret in another way.

## §2.2 Variation properties of BM

Fix  $N, t$ ,  $\Delta t = t/N$ ,  $t_i = i \Delta t$

$$\Delta W_i = W_{t_{i+1}} - W_{t_i}, W_t - W_0 = \sum \Delta W_i$$



$$\text{Now } \mathbb{E}|\Delta W_i| = \sqrt{\frac{2\Delta t}{\pi}}$$

$$\text{so } \mathbb{E} \sum_{i=1}^N |\Delta W_i| = N \sqrt{\frac{2\Delta t}{\pi}} = t \sqrt{\frac{2}{\pi \Delta t}}$$

diverges as  $N \rightarrow \infty$

$$\text{But } \sum_{i=1}^{N-1} (\Delta W_i)^2 \approx N \Delta t = t$$

Partition:  $\Pi = \{t_0, t_1, \dots\}$   
 $0 = t_0 \leq t_1 \leq t_2 \leq \dots$

$t_k \rightarrow \infty$  as  $k \rightarrow \infty$

$$\text{Mesh } \|\Pi\| = \sup_i |t_{i+1} - t_i|$$

Quadratic variation of  $X$  along  $\Pi$

$$V_t^\Pi(X) = \sum_i (X_{t_{i+1}} - X_{t_i})^2$$

$X$  has finite quadratic variation

if  $\exists \langle X \rangle_t$  finite s.t.  $\forall t \geq 0$ ,

when  $\|\Pi\| \rightarrow 0$ ,  $V_t^\Pi(X) \rightarrow \langle X \rangle_t$   
in prob. i.e.  $\forall \varepsilon, \eta > 0, \exists \delta > 0$

$$\text{s.t. } \|\Pi\| \leq \delta \Rightarrow \mathbb{P}(|V_t^\Pi(X) - \langle X \rangle_t| > \eta) < \varepsilon$$

Thm: B.M. has finite q.v.

and  $\langle W \rangle_t = t$ ,

moreover  $V_t^\pi(W) \rightarrow t$   
in  $L^2(\mathbb{P})$  and in  $\mathbb{P}$

Proof Show  $\ln L^2(\mathbb{P}) \Rightarrow \ln \mathbb{P}$ .

$\pi = \{t_0, t_1, \dots, t_n\}$ ,  $t_n = t$ .

$\Delta t_i^\pi := t_{i+1} - t_i$ ,  $\Delta W_i = W_{t_{i+1}} - W_{t_i}$

$$E[(V_t^\pi(W) - t)^2] = E\left[\left(\sum_i ((\Delta W_i)^2 - \Delta t_i^\pi)\right)^2\right]$$

$$= E\left[\sum_i ((\Delta W_i)^2 - \Delta t_i^\pi)^2\right]$$

indep. r.v.s  
mean 0  
 $\Rightarrow$  cross terms  
0 expectation

$$= \sum E[(\Delta W_i)^4 - 2\Delta t_i (\Delta W_i)^2 + (\Delta t_i)^2]$$

$$= -\sum (\Delta t_i)^2 + \sum E((\Delta W_i)^4)$$

$3(\Delta t_i)^2$

$$= 2 \sum (\Delta t_i)^2$$

$$\leq 2 \underbrace{\max_i (\Delta t_i)}_{= \|\pi\|} \cdot \underbrace{\sum \Delta t_i}_{\leq t} = 2t \|\pi\| \rightarrow 0 \quad \square$$